

Rahul Tammalla

📍 Postdam, New York | ✉ tammalr@clarkson.edu | 📞 +1 (315) 603 7792 | 🌐 Rahul.portfolio

in Rahul.linkedin | 🐙 Rahul.github

Experience

Research assistant, Clarkson University, – Potsdam, United States. Jan 2025 – Present

- Analyzed 30 million healthcare records on **Nemsis** dataset to optimize medical response times using data analysis and machine learning techniques.
- Conducted **data preprocessing** (cleaning, transforming, handling missing values) to ensure data quality and integrity across datasets.
- Performed feature engineering, statistical analysis, and hypothesis testing to identify key predictive factors affecting on-scene response time less than **10 mins**.
- Using **PostgreSQL** for database management, query optimization, and developing interactive Tableau visualizations to present actionable insights and improve resource allocation.

Associate Engineer, Cognizant Technology Solution, – Hyderabad, India. Oct 2021 – Aug 2024

- Optimized and engineered a high performance cloud-based Appian application for **Gilead Life Sciences**, a leading pharmaceutical company, processing **1M+** transactions to enhance reliability and user experience.
- Designed interactive dashboards for real-time insights, improving decision-making and visibility into key metrics.
- Improved Database and **Store procedure** performance, increasing execution speed and response time by **30%**, and ensured data integrity through daily system file analysis.
- Performed a daily in-depth data analysis on **500+** daily system records (transactions, inventory, on-hand, delivery) to ensure accuracy and seamless operations.

Education

Clarkson University, Masters in Applied Data Science (GPA : 3.889) Aug 2024 – Dec 2025

- **Coursework:** Probability Statistics, Information visualization, Database modelling, Machine Learning, Data Mining, Data warehousing.

Amrita Vishwa Vidyapeetham, Bachelors in Computer Science Engineering. May 2017 – June 2021

Projects

AI Research Paper Summarization [pdfplumber, scikit-learn, Hugging Face Transformers (T5), PyTorch, Git, Streamlit] 2025

- Developed application that **summarizes academic research papers** by extracting text from PDFs using **pdfplumber** and generating **extractive summary** using **TextRank**, producing human-like **abstractive summary** with **Hugging Face T5 LLM model**, and evaluating summary quality using **ROUGE-1** and **ROUGE-2** metrics.

News Scraper [Transformers, Pandas, NewsAPI, DistilBERT, T5-small, Streamlit] 2025

- Developed app that scrapes top US news using NewsAPI, performs **sentiment analysis** using **DistilBERT**, and **summarizes articles** using **T5-small LLM model**, updated the news every 24 hours.

Medical Chat Bot [Python, Neo4j, Cypher Queries, RAG, GPT, spaCy, Streamlit] 2025

- Developed a **hybrid chatbot** that integrates **Neo4j Knowledge Graphs** with **GPT** using **Retrieval-Augmented Generation (RAG)** to ensure **0% hallucination** for structured queries while handling general medical queries.

Snow Depth Forecasting [ARIMA, AutoARIMA, HoltWinters] 2025

- Designed and implemented time series forecasting **machine learning** models to predict snow depth in Tupper Lake for 2026, and **optimized** accuracy by **model evaluation**.

US Accident Insights [dplyr, lubridate, ggplot2, RShiny] 2024

- Developed the "US Accident Insights" **Shiny app**, analyzing **40K+** records of accidents across United States and visualizing traffic accident data using **R libraries** with **Geo-spatial** visualizations.

Personalized Course Recommendation Database System [SQL] 2024

- Developed an dynamic course recommendation system with **SQL triggers** and **procedures** to optimized databases for exam schedules, instructor workloads, and student performance.

Skills

Programming & Development: Python, SQL, R, Java, MATLAB, Agile, Scrum, OOP , HTML & CSS , MS tools.

Libraries & Frameworks: NumPy, Pandas, SciPy, scikit-learn, PySpark, TensorFlow, Keras, OpenCV, Matplotlib, Seaborn, AWS, PyTorch ,Azure.

Data Engineering & Visualization: ETL, CI/CD, DevOps,R Studio, Tableau, Power BI.

Databases & Cloud: PostgreSQL, MongoDB, Snowflake, Hadoop, Hive, Streamlit, Neo4j.

Machine Learning : Regression, Classification, Clustering, Random Forests, XGBoost, SVM, Statistical Modeling, Ensemble Learning, CNN, Neural Networks, Time series.

NLP : Hugging Face Transformers-(T5,DistilBERT),spaCy, RAG(Retrieval-Augmented Generation).

Certifications

Advanced NLP with Python for Machine Learning - [Linkedin learning]	2025
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Hands-On PyTorch Machine Learning - [Linkedin learning]	2025
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